



WeCloudData

Wildlife Reserve Control & Detection System

Identifying species in camera-trap images
from cameras the model has never seen

Technical Report · Group 10

TEAM

Abdulrahman Bin Shannar
Mansour Almutairi
Muhannad Qahl

PROGRAMME

WeCloudData Data Science Bootcamp
Saudi Digital Academy

CLIENT CONTEXT

King Abdulaziz Royal Reserve (KARR)

DATASET

iWildCam 2020 (WILDS v2.0)

DATE

September 2026





Executive Summary

The project on one page

The King Abdulaziz Royal Reserve (KARR) runs 103 camera traps. Every trigger produces a burst of photos, most of them empty or near-identical, and each still needs a person to review it. We asked one question: **can a model name the species in a camera-trap photo taken by a camera it has never seen?** Using the public iWildCam 2020 benchmark as a stand-in for KARR data, we explored all 203,029 images, modelled the ten most frequent classes, and scored every model on **29 camera locations never used in training**.

0.908

honest macro F1 per burst,
final model

90.3%

burst accuracy,
final model

0.716

best per-burst score
from our data alone

-90%

estimated review hours
at KARR scale

- **Final model:** MegaDetector finds the animal, SpeciesNet describes the crop, a small MLP maps that to our 10 classes, and the most confident frame decides each burst.
- **Clean baseline:** SpeciesNet may have seen iWildCam images in its own training, so we also report our best model trained only on our data (BioCLIP crops + logistic regression / XGBoost): 0.598 per image, 0.716 per burst.
- **What moved the score:** better inputs, not tuning. Pretraining +0.10, cropping the animal +0.11 and a camera-trap foundation model +0.14; a full round of CNN tuning added only +0.025.
- **Operational impact:** about 1.01 million images a year at KARR scale. One decision per burst plus a blank filter cuts estimated review from 844 to 88 hours a year, and edge data by about 88%.

RECOMMENDATION Build on camera-trap foundation models with a detect-then-classify pipeline, add a whole-image fallback for frames without a confident detection, and fine-tune on KARR's own labelled species before deployment.



1 Problem

What problem were we trying to solve?

KARR monitors wildlife with 103 field cameras across about 28,000 km². Camera traps fire on motion or heat, so each event becomes a **burst** of three to ten near-identical frames, and many events contain no animal at all. Today every image goes through a manual review queue. That costs **reviewer hours**, **delays** rare-species sightings, gives no signal for **relocating** low-yield cameras, and wastes **bandwidth** on blank frames sent over cellular links.

OUR QUESTION Can a model identify the species in a camera-trap image taken by a camera location it was never trained on? New cameras are installed all the time, so a model that only works on familiar backgrounds is of no use in the field.

Objectives: filter empty frames; classify species with per-species precision and recall; generalise to unseen cameras; and estimate the operational savings. iWildCam is a declared stand-in: forest and savanna cameras rather than Saudi desert, so every KARR figure is a planning estimate.



2 Data

What data did we use?

iWildCam 2020, WILDS v2.0 release: 203,029 labelled images from 323 camera locations, 182 classes and 36,484 bursts (2013–2015). The metadata is clean, with 0 missing cells and 0 duplicate IDs; a 99999 placeholder label in the category file was filtered at load. The release has **no GPS coordinates and no bounding boxes**. WILDS splits the data by **camera**: the OOD splits share no cameras with training, so we used the official splits and never shuffled.

We modelled the **ten most frequent classes**: empty plus ocellated turkey, great curassow, white-lipped peccary, black-fronted duiker, agouti, ocelot, red brocket deer, puma and gray fox. Each class was capped at 3,000 training and 600 evaluation images, sampled across bursts, with the same seed in every notebook.

Table 1. Modelling sample (39,898 images).

Split	Role	Images	Bursts	Cameras
train	Fitting (25,211 images) + model selection (4,529 images, 36 held-out cameras)	29,740	8,773	208
id_val / id_test	Seen cameras, later dates	2,377 / 3,420	872 / 982	107 / 137
ood_val	Never-seen cameras: the real test	4,361	1,444	29



3 Analysis

What did we learn from the data?

Our EDA covered the full release before any sampling. Each finding led to a design decision (summarised in Table 3).

34.2%

of frames are blank

5.56

frames per burst

46%

of frames are infrared

1.6%

of the frame: median animal

3.1 Most of the volume is waste

At 26.9 images per camera-day, KARR's 103 cameras would produce about **1.01 million images a year**. 34.2% of frames are blank, and because bursts average 5.56 frames, 82% of frames repeat the first one in their burst. **43% of bursts are blank end to end** and only 5% are mixed, so a burst is essentially one decision. At 3 s per decision, review falls from **844 h** a year (every frame) to **152 h** (one decision per burst) and **88 h** with a blank filter; sending one frame per burst cuts edge data from about 44 to 5 GB a year.

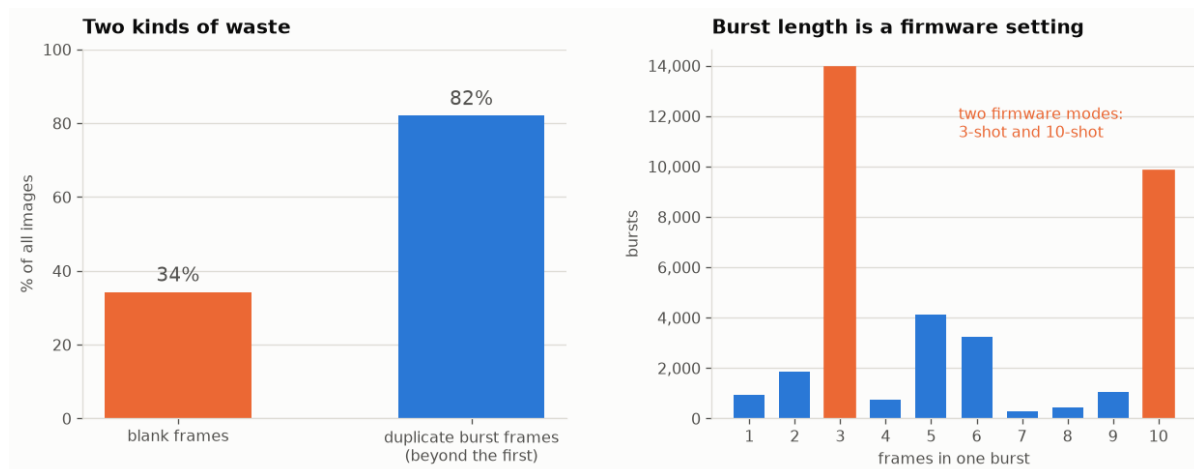


Figure 1. Two kinds of waste (left); burst length spikes at 3 and 10, which are camera firmware settings (right). Source: EDA notebook.

3.2 A brutal long tail

There are 181 species plus empty. Ocellated turkey alone has 16,282 images (8% of the dataset), while **54 species have ten photos or fewer**. The top ten classes hold 56% of animal photos, and only 102 of 182 classes appear in the official OOD test split. Accuracy is therefore misleading: the published WILDS baseline scores 71.6% accuracy but only 31.0 macro F1 on the same predictions. We use macro F1 and capped, class-balanced training, and scoped modelling to the top ten classes.

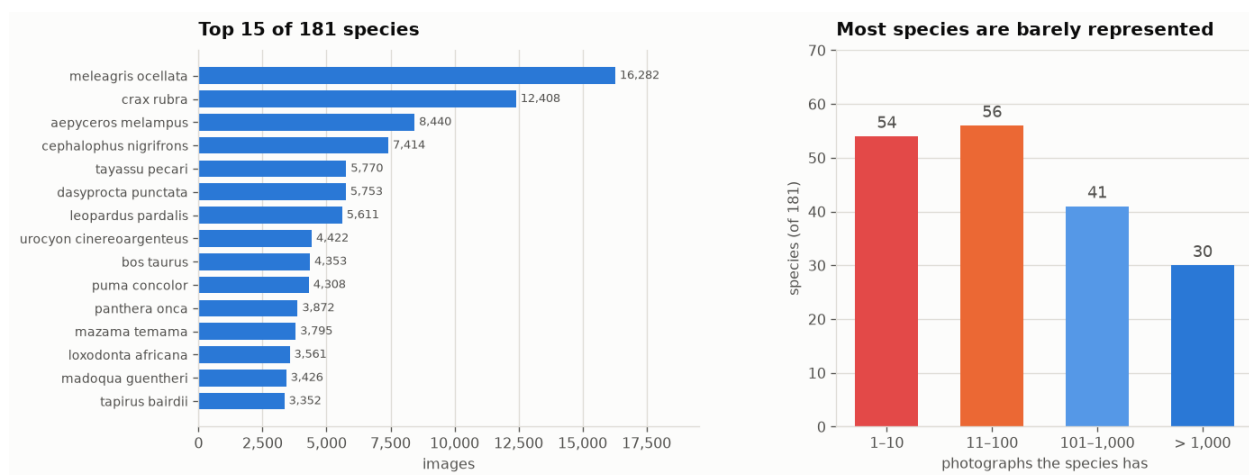


Figure 2. Top 15 species (left) and species grouped by photo count (right). Source: EDA notebook.

3.3 Time, light and camera fingerprints

Splitting the day three ways separates species well (81 diurnal, 38 crepuscular, 62 nocturnal), so hour of day is a useful feature. The calendar is not: 92% of images are from 2013, so month and year only encode which cameras were running. 46% of frames are infrared, including 30% of daytime frames under dense canopy. And 45% of frames carry a burned-in banner (camera model, sometimes an ID) that can let a network recognise the camera.

3.4 The animal is tiny, and new cameras see different animals

With no boxes, background subtraction puts the **median animal at 1.6% of the frame** (about 28×28 px at 224 px input), with 42% of bursts under 1%: a whole-frame classifier mostly sees scenery. Comparing training cameras with OOD test cameras, the largest shift is **which species appear** (distance 0.237), ahead of hour of day (0.169) and infrared share (0.096).

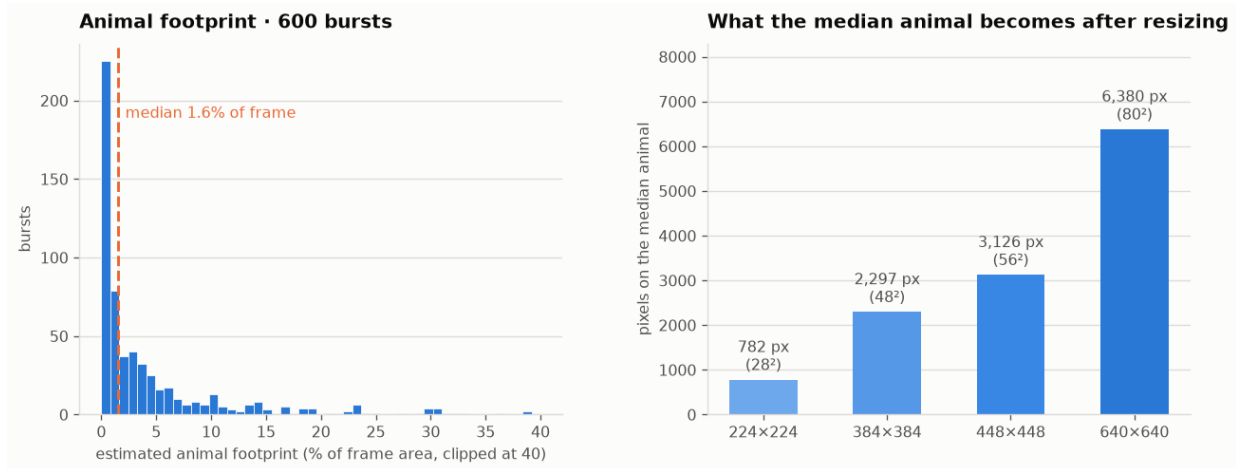


Figure 3. Estimated animal footprint in 600 bursts (left) and the pixels the median animal keeps at common input sizes (right). Source: EDA notebook.

Table 2. From EDA findings to design decisions.

EDA finding	Design decision	Status
Splits are cut by camera	Official splits only; never shuffle	Adopted
Long tail; accuracy misleads	Macro F1 and an 'honest' variant (Section 4)	Adopted
Extreme class imbalance	Cap each class; class-balanced weights	Adopted
Bursts of near-identical frames	One answer per burst	Adopted
Median animal is 1.6% of the frame	Detect first, classify the crop	Adopted
Calendar encodes the campaign	No month or year features; keep hour of day	Adopted
46% of frames are infrared	Greyscale augmentation; evaluate IR separately	Not yet
Banner names the camera	Crop top and bottom 5% of each frame	Not yet



4 Approach

How did we solve the problem?

Three evaluation rules keep the numbers honest about new cameras:

- 1 Split by camera.** Train on training cameras; test on the 29 OOD cameras (4,361 images) never seen in training.
- 2 Select without peeking.** Choose between models on 36 held-out training cameras; read the test split once, afterwards.
- 3 Honest macro F1.** Black-fronted duiker and empty appear at cameras no other class visits, so a model can score them from the background alone. Honest macro F1 averages F1 over the other eight classes. We report it per image and per burst.

All final models share one pipeline:

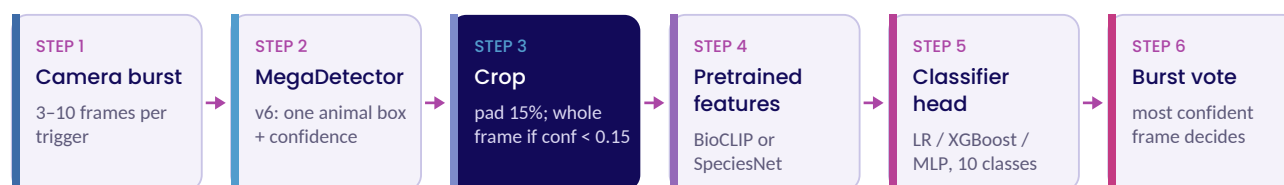


Figure 4. Detect → crop → classify → vote.

MegaDetector v6 was validated against labels before we used it for cropping: its confidence separates animal from empty frames with an **AUC of 0.82** on unseen cameras. The median crop is half the frame's width, a 2× zoom on the typical animal. An earlier hand-made background-subtraction detector produced boxes covering 93% of the frame, which is why a trained detector replaced it. Tabular features (hour of day and box geometry) were each audited for how much of their variance is explained by the camera, and dropped if above 50%.



5 Models

What models did we try?

Five model families, built in sequence, each fixing the weakness the last one exposed:

Table 3. Model families (*honest macro F1 on the 29 unseen cameras, per image).

Model	What it is	Honest F1*
CNN from scratch	Small CNN trained only on our photos (64–128 px), as proposed. Less overfitting helped seen cameras but not unseen ones: it learned backgrounds.	0.320
ResNet-18 fine-tuned	ImageNet-pretrained, 192 px, layer-wise learning rates, mixup, label smoothing, AdamW + cosine schedule, 12 epochs.	0.446
EfficientNetV2-S + XGBoost	Pretrained image embeddings fed to a gradient-boosted tree classifier.	0.437
BioCLIP + LR / XGBoost	Biology-specific CLIP features of the MegaDetector crop, PCA to 128 dims plus 7 tabular features; logistic regression, XGBoost and soft votes.	0.598
SpeciesNet + MLP	Google's camera-trap classifier (EfficientNetV2-M, ~65M images) on the crop; an MLP (471 → 256 → 10) maps its label scores to our classes.	0.734

Model selection. For BioCLIP, 13 candidates (three feature streams × logistic regression / XGBoost, plus votes) were compared on the selection cameras; the one-standard-error rule picked **vote (crop)**. For SpeciesNet, ten candidates (zero-shot, probes and averages, each with two label-mapping variants) were compared; the **probe**

on SpeciesNet won with 0.954 burst honest F1. The selection cameras ranked candidates the same way as the test cameras for 84% of pairs, so the selection was reliable.



6 Comparison

How did the models perform?

Table 4. All approaches on the 29 unseen cameras ('-' = not reported in that notebook).

Approach	Accuracy	Macro F1	Honest F1	Honest F1 / burst
CNN from scratch (64 px)	-	0.434	0.320	-
ResNet-18 v3 (192 px, layer-wise LR, mixup)	0.637	0.543	0.446	-
EfficientNetV2-S + XGBoost	0.581	0.512	0.437	-
BioCLIP, full frame	0.653	0.581	0.490	-
BioCLIP + MegaDetector crop (LR + XGBoost vote)	0.725	0.662	0.598	0.716
SpeciesNet zero-shot (no training)	-	-	-	0.825
SpeciesNet + MLP probe (final)	0.752	0.734	0.734	0.908

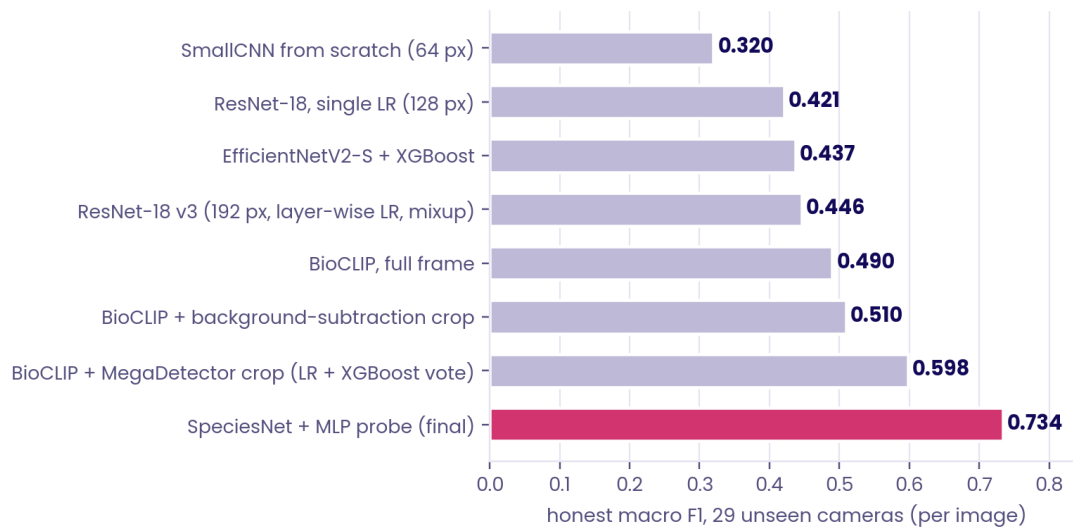


Figure 5. Honest macro F1 on unseen cameras, per image.

The gains came in three steps, none of them tuning: **pretraining** +0.10, **cropping the animal** +0.11 and a **camera-trap foundation model** +0.14. Everything in ResNet-18 v3 added only +0.025. Voting once per burst adds a further +0.12 to +0.17.

Table 5. Logistic regression vs XGBoost (honest F1): the input matters far more than the classifier.

Same BioCLIP inputs	Logistic regression	XGBoost
Full frame + tabular	0.506	0.493
Crop + tabular	0.580	0.582

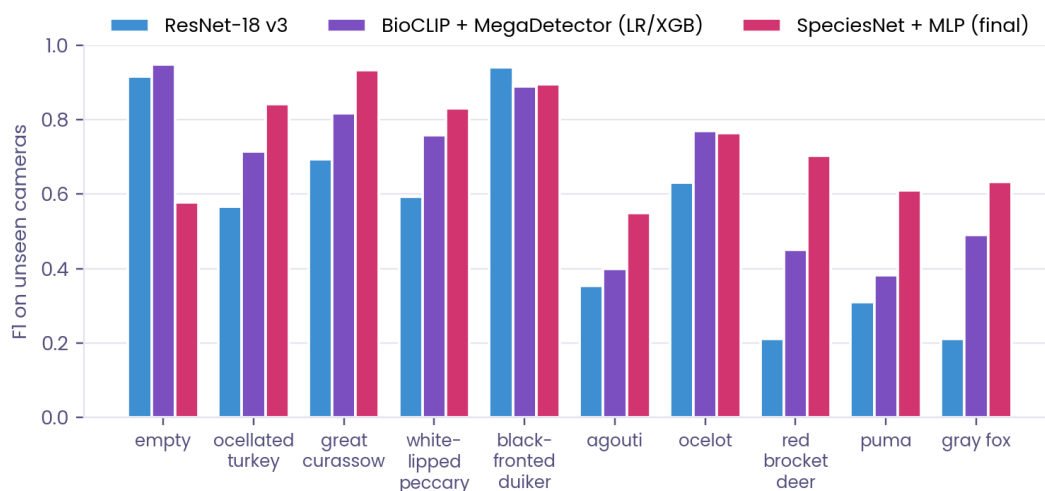


Figure 6. F1 per class on unseen cameras for three generations of model (final model's values are in Table 7).



7 Final Model

Which model did we select, and why?

SpeciesNet + MLP probe, voted per burst. MegaDetector v6 finds the animal; the padded crop (480 px) goes to SpeciesNet, whose 461 label scores plus 10 mapped class scores form a 471-value input to a small MLP (256 hidden units, dropout 0.2, class-weighted loss); the most confident frame in each burst decides the answer.

- It won selection on the held-out cameras, chosen before the test set was read.
- It beats our best data-only pipeline by +0.19 honest F1 per burst (0.908 vs 0.716).
- It improves most where earlier models failed: red brocket deer 0.45 → 0.70, puma 0.38 → 0.61, gray fox 0.49 → 0.63.
- It is small and deployable: a 484 KB head on top of the public SpeciesNet model, already exported for the web model server.

Table 6. Final model on the 29 unseen cameras.

View	Accuracy	Macro F1	Honest macro F1
Per image	0.752	0.734	0.734
Per burst	0.903	0.901	0.908

Table 7. Final model per-class results on unseen cameras (per image).

Class	Precision	Recall	F1	Support
empty (false trigger)	0.409	0.983	0.578	600
ocellated turkey	0.982	0.737	0.842	600
great curassow	0.983	0.890	0.934	600
white-lipped peccary	0.901	0.772	0.831	600
black-fronted duiker	0.996	0.815	0.896	600
agouti	0.813	0.414	0.549	210
ocelot	0.980	0.628	0.765	470
red brocket deer	0.810	0.623	0.704	130
puma	0.623	0.598	0.610	241
gray fox	0.857	0.503	0.634	310

Where it fails. Empty has recall 0.98 but precision 0.41: **28% of animal crops get 'blank' as SpeciesNet's top label.** A label-mapping bug was ruled out (only 0.3% of images have no mapped label). The likely cause is the ~18–22% of frames where MegaDetector found no confident box, so the whole frame went to a crop classifier. Agouti, puma and gray fox, the smallest training classes, remain the weakest species.

CAVEAT SpeciesNet was trained on public camera-trap repositories, and iWildCam is public, so it may have seen these images. The final score shows what the best available model can do; the BioCLIP pipeline (0.598 per image, 0.716 per burst) is the clean measure of what can be learned from this training set alone.

8 Results & Insights

What did we learn?

- 1 Pretrained features are the biggest lever:** 0.320 (scratch) → 0.421 (ImageNet) → 0.734 (camera-trap model).
- 2 Tuning cannot fix a shortcut:** mixup and layer-wise rates cut the train/val gap (0.215 → 0.144) but moved unseen-camera F1 by only +0.025.
- 3 Look at the animal, not the scene:** detector crops lifted BioCLIP from 0.490 to 0.598.
- 4 One trigger, one vote:** deciding per burst added +0.118 and +0.174 to the two best pipelines.
- 5 The classifier matters less than its inputs:** logistic regression 0.580 vs XGBoost 0.582 on the same crops.
- 6 Domain models work out of the box:** SpeciesNet zero-shot reached 0.825 per burst with no training.
- 7 Honest evaluation changes the story:** the same ResNet predictions score 0.637 image-weighted but 0.807 camera-weighted; at the 6 cameras with several species it drops to 53% accuracy.

Table 8. Operational impact, extrapolated from iWildCam to KARR's 103 cameras.

For KARR (planning estimates)	Now	With the system
Manual review per year (3 s per decision)	≈844 h	≈88 h (–90%)
Data sent over cellular per year	≈44 GB	≈5 GB (–88%)
Rare-species alerts	none	top-k per day, ranked by per-station rarity



9 Conclusion

What are the main takeaways?

On 29 camera locations it never saw, our final pipeline reaches **0.908 honest macro F1 and 90.3% accuracy per burst**; the best model learned from our data alone reaches 0.716. Both come from an evaluation built to resist the location shortcut, and the remaining errors sit in two known places: animals labelled empty, and the rarest small mammals.

- Generalising to new cameras works only when the model looks at the animal, not the scene.
- For a new reserve, start from camera-trap foundation models rather than training from scratch.
- Most of the operational saving comes from structure (one decision per burst, a blank filter) before any species model runs.
- Honest evaluation (camera splits, held-out selection cameras, the honest metric) is what makes the numbers trustworthy.

Limitations

iWildCam is forest and savanna, not Saudi desert, so KARR figures are planning estimates. We modelled 10 of 182 classes on the OOD validation cameras, so scores are not comparable to the WILDS leaderboard; SpeciesNet may have seen these images; and some per-class scores rest on 130–310 test images.

Next steps

- 1 Add SpeciesNet's whole-image classifier for frames with no confident detection, to fix the empty-class errors.
- 2 Fine-tune on KARR's own labelled species (for example the Arabian gazelle) before field deployment.
- 3 Crop the burned-in banners, add greyscale augmentation, and report infrared and colour frames separately.
- 4 Serve the exported model behind a capacity-budgeted review queue and a per-station dashboard.



Appendix

Glossary and references

Term	Meaning
Burst	The photos one trigger produces, usually 3–10 frames in quick succession (seq_id).
Empty / blank	A frame with no animal, usually triggered by wind, vegetation or heat.
OOD split	Photos from cameras never seen in training (out of distribution).
Honest macro F1	F1 per class averaged over the 8 classes that require seeing the animal (excludes black-fronted duiker and empty).
MegaDetector / BioCLIP / SpeciesNet	Pretrained models that find animals / describe biological images / classify camera-trap species.
Probe	A small classifier trained on top of a frozen pretrained model's outputs.

References. Beery, Cole & Gjoka (2020), The iWildCam 2020 Competition Dataset, arXiv:2004.10340 · Koh, Sagawa et al. (2021), WILDS: A Benchmark of in-the-Wild Distribution Shifts, ICML · Beery, Morris & Yang (2019), Efficient Pipeline for Camera Trap Image Review (MegaDetector), arXiv:1907.06772 · Stevens et al. (2024), BioCLIP, CVPR · Gadot et al. (2024), To crop or not to crop (SpeciesNet), IET Computer Vision.